Standard Model

1. Spontaneous Symmetry Breaking

Spontaneous symmetry breaking is the situation in which the equations and the Lagrangian have a symmetry, but the vacuum chosen by the theory does not. The symmetry is still present in the theory; it is not a mistake in the Lagrangian. The point is that the set of lowest-energy field configurations contains more than one equivalent choice, and choosing one of them hides part of the symmetry.

This section develops the idea in three steps. First we study a real scalar with a discrete symmetry, where there is symmetry breaking but no Goldstone boson. Then we study a complex scalar with a continuous $U(1)$ symmetry, where a massless Goldstone mode appears. Finally we generalize to N complex scalars and prove the Goldstone theorem by counting flat directions of the potential.

1.1 A real scalar with a discrete symmetry

Consider one real scalar field ϕ with Lagrangian

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - V(\phi), \tag{4.1}$$

and potential

$$V(\phi) = -\frac{1}{2}\mu^2\phi^2 + \frac{\lambda}{4}\phi^4, \qquad \mu^2 > 0, \quad \lambda > 0. \tag{4.2}$$

This is often called the "wrong-sign mass" potential. Around $\phi = 0$ the quadratic term in the potential has negative curvature, so $\phi = 0$ is not a stable vacuum. This corrects a common misleading phrase: the theory does have a quadratic term; what is wrong is the sign of the quadratic term around the origin.

The potential is invariant under the discrete transformation

$$\phi \longrightarrow -\phi, \qquad V(\phi) = V(-\phi). \tag{4.3}$$

This is a $\mathbb{Z}_2$ symmetry.

To find the extrema, compute

$$\frac{\mathrm{d}V}{\mathrm{d}\phi} = -\mu^2\phi + \lambda\phi^3 = \phi\left(-\mu^2 + \lambda\phi^2\right). \tag{4.4}$$

Thus

$$\frac{\mathrm{d}V}{\mathrm{d}\phi} = 0 \qquad \Longrightarrow \qquad \phi = 0 \quad \text{or} \quad \phi^2 = \frac{\mu^2}{\lambda}. \tag{4.5}$$

Define

$$v = \frac{\mu}{\sqrt{\lambda}}. \tag{4.6}$$

The extrema are therefore

$$\phi = 0, \qquad \phi = \pm v. \tag{4.7}$$

To decide which are minima, compute the second derivative:

$$\frac{\mathrm{d}^2 V}{\mathrm{d}\phi^2} = -\mu^2 + 3\lambda\phi^2. \tag{4.8}$$

At the origin,

$$\left.\frac{\mathrm{d}^2 V}{\mathrm{d}\phi^2}\right|_{\phi=0} = -\mu^2 < 0, \tag{4.9}$$

so $\phi = 0$ is a local maximum. At $\phi = \pm v$,

$$\left.\frac{\mathrm{d}^2 V}{\mathrm{d}\phi^2}\right|_{\phi=\pm v} = -\mu^2 + 3\lambda\frac{\mu^2}{\lambda} = 2\mu^2 > 0, \tag{4.10}$$

so $\phi = \pm v$ are true minima.

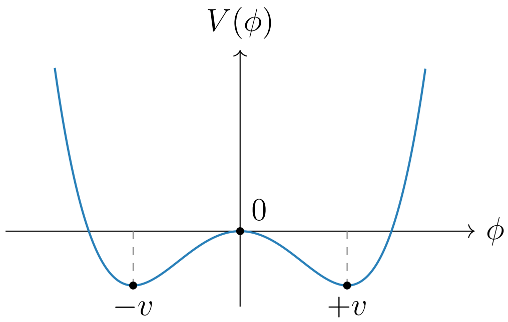


Figure 4.1: The double-well potential. The Lagrangian is symmetric under $\phi \to -\phi$, but either vacuum $\phi = +v$ or $\phi = -v$ chooses one side.

The theory has the symmetry $\phi \to -\phi$, but the vacuum does not. If the system chooses

$$\langle 0|\phi|0\rangle = +v, \tag{4.11}$$

then the transformed configuration has

$$\langle 0|\phi|0\rangle \longrightarrow -v, \tag{4.12}$$

which is a different vacuum. The two vacua are physically equivalent, but once one is chosen the symmetry is no longer manifest. This is spontaneous breaking of a discrete symmetry.

There is no Goldstone boson in this example. Goldstone bosons arise from broken continuous symmetries, where moving along the vacuum manifold can be done infinitesimally with no energy cost. A discrete symmetry does not provide such a continuous flat direction.

1.2 A complex scalar and global U(1) breaking

Now consider a complex scalar field

$$\phi = \frac{1}{\sqrt{2}}\left(\phi_1 + \mathrm{i}\phi_2\right), \tag{4.13}$$

with Lagrangian

$$\mathcal{L} = \partial_\mu\phi^*\,\partial^\mu\phi - V(\phi, \phi^*), \tag{4.14}$$

and potential

$$V(\phi,\phi^*) = -\mu^2\phi^*\phi + \frac{\lambda}{2}\left(\phi^*\phi\right)^2, \qquad \mu^2>0, \quad \lambda>0. \tag{4.15}$$

The Lagrangian is invariant under the global $U(1)$ transformation

$$\phi(x) \longrightarrow \phi'(x) = e^{\mathrm{i}\alpha}\phi(x), \qquad \alpha\in\mathbb{R}. \tag{4.16}$$

This is a global symmetry because α is constant in spacetime.

The potential depends only on the radial variable $\phi^*\phi$. Minimizing with respect to ϕ^* gives

$$\frac{\partial V}{\partial\phi^*} = \phi\left(-\mu^2+\lambda\phi^*\phi\right) = 0. \tag{4.17}$$

Therefore

$$\phi = 0 \quad \text{or} \quad \phi^*\phi = \frac{\mu^2}{\lambda}. \tag{4.18}$$

Again $\phi=0$ is the unstable point. The true vacua satisfy

$$|\phi|^2 = \frac{\mu^2}{\lambda}. \tag{4.19}$$

In the complex plane this is a circle. Since

$$|\phi|^2 = \frac{1}{2}\left(\phi_1^2+\phi_2^2\right), \tag{4.20}$$

the vacuum circle can also be written as

$$\phi_1^2+\phi_2^2 = \frac{2\mu^2}{\lambda}. \tag{4.21}$$

It is conventional to define

$$v^2 = \frac{2\mu^2}{\lambda}. \tag{4.22}$$

Then the vacuum condition is

$$|\phi|^2 = \frac{v^2}{2}. \tag{4.23}$$

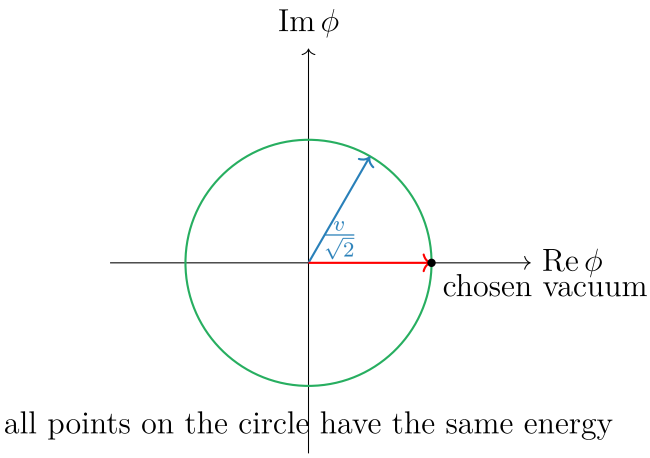


Figure 4.2: The vacuum manifold for the broken global $U(1)$ model. Choosing one point on the circle breaks the manifest $U(1)$ symmetry.

Every point on this circle is a possible ground state. Choosing one point breaks the global $U(1)$ symmetry. We may choose the vacuum to lie along the real direction:

$$\phi_0 = \frac{v}{\sqrt{2}}, \qquad \phi_{1,0} = v, \quad \phi_{2,0} = 0. \tag{4.24}$$

This choice is not a loss of generality; any other point on the circle can be rotated into this one by a $U(1)$ transformation.

1.3 Expansion around the chosen vacuum

To find the physical particles, expand the field around the chosen vacuum:

$$\phi(x) = \frac{1}{\sqrt{2}} \left[v + h(x) + \mathrm{i}\sigma(x)\right]. \tag{4.25}$$

Here $h(x)$ is the radial fluctuation and $\sigma(x)$ is the angular fluctuation. At the vacuum,

$$h = 0, \qquad \sigma = 0. \tag{4.26}$$

The kinetic term becomes

$$\begin{aligned}\partial_\mu \phi^* \, \partial^\mu \phi &= \frac{1}{2} \partial_\mu (v + h - \mathrm{i}\sigma) \partial^\mu (v + h + \mathrm{i}\sigma) \\ &= \frac{1}{2} \partial_\mu h \, \partial^\mu h + \frac{1}{2} \partial_\mu \sigma \, \partial^\mu \sigma.\end{aligned} \tag{4.27}$$

Since v is constant, its derivative vanishes.

Now compute the potential in terms of h and σ. First,

$$\begin{aligned}\phi^* \phi &= \frac{1}{2} \left[(v + h)^2 + \sigma^2\right] \\ &= \frac{1}{2} \left[v^2 + 2vh + h^2 + \sigma^2\right].\end{aligned} \tag{4.28}$$

Therefore

$$V(h, \sigma) = -\frac{\mu^2}{2} \left[(v + h)^2 + \sigma^2\right] + \frac{\lambda}{8} \left[(v + h)^2 + \sigma^2\right]^2. \tag{4.29}$$

Expanding the square,

$$\begin{aligned}\left[(v + h)^2 + \sigma^2\right]^2 &= \left(v^2 + 2vh + h^2 + \sigma^2\right)^2 \\ &= v^4 + 4v^3 h + 6v^2 h^2 + 2v^2 \sigma^2 + \mathcal{O}(\text{cubic}).\end{aligned} \tag{4.30}$$

Hence the terms up to quadratic order are

$$V(h, \sigma) = -\frac{\mu^2}{2} \left(v^2 + 2vh + h^2 + \sigma^2\right) + \frac{\lambda}{8} \left(v^4 + 4v^3 h + 6v^2 h^2 + 2v^2 \sigma^2\right) + \cdots. \tag{4.31}$$

Separate constant, linear, and quadratic terms:

$$\begin{aligned}V(h, \sigma) = &\left(-\frac{\mu^2 v^2}{2} + \frac{\lambda v^4}{8}\right) + \left(-\mu^2 v + \frac{\lambda v^3}{2}\right) h \\ &+ \left(-\frac{\mu^2}{2} + \frac{3\lambda v^2}{4}\right) h^2 + \left(-\frac{\mu^2}{2} + \frac{\lambda v^2}{4}\right) \sigma^2 + \cdots.\end{aligned} \tag{4.32}$$

Using

$$v^2 = \frac{2\mu^2}{\lambda}, \tag{4.33}$$

the linear term vanishes:

$$-\mu^2 v + \frac{\lambda v^3}{2} = v \left(-\mu^2 + \frac{\lambda v^2}{2}\right) = 0. \tag{4.34}$$

This must happen because we expanded around a minimum. The quadratic terms become

$$-\frac{\mu^2}{2}+\frac{3\lambda v^2}{4}=-\frac{\mu^2}{2}+\frac{3\mu^2}{2}=\mu^2, \tag{4.35}$$
$$-\frac{\mu^2}{2}+\frac{\lambda v^2}{4}=-\frac{\mu^2}{2}+\frac{\mu^2}{2}=0. \tag{4.36}$$

Thus

$$V(h,\sigma)=V(v)+\mu^2h^2+\mathcal{O}(h^3,h^2\sigma,h\sigma^2,\sigma^4). \tag{4.37}$$

The Lagrangian contains $-\mu^2h^2$. Comparing with the standard mass term

$$-\frac{1}{2}m_h^2h^2, \tag{4.38}$$

we find

$$m_h^2=2\mu^2. \tag{4.39}$$

There is no quadratic term for σ, so

$$m_\sigma^2=0. \tag{4.40}$$

The radial field h is massive. The angular field σ is massless and is called the Goldstone boson.

The reason is geometric. The h direction moves away from the vacuum circle, so the potential increases. The σ direction moves along the vacuum circle, where the potential is flat. A flat direction has zero curvature, and zero curvature means zero mass squared.

1.4 N complex scalar fields

Now consider N complex scalar fields ϕ^i, $i=1,\dots,N$, with

$$\mathcal{L}=\sum_{i=1}^{N}\partial_\mu\phi^{i*}\partial^\mu\phi^i-V(\phi,\phi^*), \tag{4.41}$$

and

$$V(\phi,\phi^*)=-\mu^2\sum_{i=1}^{N}\phi^{i*}\phi^i+\frac{\lambda}{2}\left(\sum_{i=1}^{N}\phi^{i*}\phi^i\right)^2. \tag{4.42}$$

This potential is invariant under global $U(N)$ transformations:

$$\phi^i\longrightarrow\phi'^i=\sum_{j=1}^{N}U^{ij}\phi^j,\qquad U^\dagger U=\mathbf{1}. \tag{4.43}$$

The quantity

$$\sum_i\phi^{i*}\phi^i \tag{4.44}$$

is just the squared length of the complex vector ϕ, so it is unchanged by unitary rotations.

The minimum satisfies

$$\sum_{i=1}^{N}\phi^{i*}\phi^i=\frac{\mu^2}{\lambda}. \tag{4.45}$$

This is a sphere in the $2N$ real-dimensional field space. Define again

$$\frac{v^2}{2}=\frac{\mu^2}{\lambda},\qquad v^2=\frac{2\mu^2}{\lambda}. \tag{4.46}$$

Using the $U(N)$ symmetry, choose the vacuum to point in the first complex direction:

$$\phi_0^1 = \frac{v}{\sqrt{2}}, \qquad \phi_0^2 = \cdots = \phi_0^N = 0. \tag{4.47}$$

Then write the fields as

$$\phi^1 = \frac{1}{\sqrt{2}} \left[v + h(x) + \mathrm{i}\sigma(x)\right], \tag{4.48}$$

$$\phi^k = \frac{1}{\sqrt{2}} \left[\pi_{2k-2}(x) + \mathrm{i}\pi_{2k-1}(x)\right], \qquad k = 2, \dots, N. \tag{4.49}$$

There are $2N$ real fields in total:

$$h, \quad \sigma, \quad \pi_2, \dots, \pi_{2N-1}. \tag{4.50}$$

The radial field h changes the length of the vector ϕ and is massive. All other fields change only the direction of ϕ in the vacuum manifold and are massless at tree level.

Equivalently, the chosen vacuum leaves a residual $U(N-1)$ symmetry acting on the fields

$$\phi^2, \dots, \phi^N. \tag{4.51}$$

Thus the symmetry-breaking pattern is

$$U(N) \longrightarrow U(N-1). \tag{4.52}$$

The number of broken generators is

$$\dim U(N) - \dim U(N-1) = N^2 - (N-1)^2 = 2N - 1. \tag{4.53}$$

Therefore there are

$$2N - 1 \tag{4.54}$$

massless Goldstone bosons. This agrees with the field decomposition above: σ plus the $2N-2$ real components of $\phi^2, \dots, \phi^N$.

1.5 Goldstone theorem

The examples above illustrate the general result.

> **Goldstone theorem.** In a relativistic theory with a continuous global internal symmetry, each spontaneously broken generator gives one massless real scalar excitation, called a Goldstone boson.

There are important assumptions in this statement. The symmetry must be global, not gauged. If the symmetry is gauged, the massless scalar is removed from the physical spectrum by the Higgs mechanism. Also, the symmetry must be exact. If it is explicitly broken, the would-be Goldstone boson generally becomes light but not massless.

1.6 Proof by the mass matrix

Consider N real scalar fields ϕ^i with Lagrangian

$$\mathcal{L} = \frac{1}{2}\sum_{i=1}^{N} \partial_\mu \phi^i \, \partial^\mu \phi^i - V(\phi). \tag{4.55}$$

Suppose the potential is invariant under a continuous symmetry group G. For an infinitesimal transformation,

$$\phi^i \longrightarrow \phi^i + \delta_a \phi^i, \tag{4.56}$$

where

$$\delta_a \phi^i = \alpha^a (T_a)^i{}_j \phi^j. \tag{4.57}$$

Here T_a is a generator of the symmetry and α^a is an infinitesimal real parameter. Since V is invariant,

$$\delta_a V = \sum_i \frac{\partial V}{\partial \phi^i} \delta_a \phi^i = 0. \tag{4.58}$$

Substitute the infinitesimal transformation:

$$\sum_i \frac{\partial V}{\partial \phi^i} (T_a)^i{}_j \phi^j = 0. \tag{4.59}$$

This identity holds for every field configuration ϕ, not only at the minimum.

Now differentiate this identity with respect to ϕ^k:

$$\sum_i \frac{\partial^2 V}{\partial \phi^k \partial \phi^i} (T_a)^i{}_j \phi^j + \sum_i \frac{\partial V}{\partial \phi^i} (T_a)^i{}_k = 0. \tag{4.60}$$

Evaluate at a vacuum ϕ_0. Since ϕ_0 is a minimum,

$$\left. \frac{\partial V}{\partial \phi^i} \right|_{\phi_0} = 0. \tag{4.61}$$

The second term therefore vanishes. Define the mass matrix

$$(M^2)_{ki} = \left. \frac{\partial^2 V}{\partial \phi^k \partial \phi^i} \right|_{\phi_0}. \tag{4.62}$$

Then

$$\sum_i (M^2)_{ki} (T_a)^i{}_j \phi_0^j = 0. \tag{4.63}$$

In vector notation this is

$$M^2 \, (T_a \phi_0) = 0. \tag{4.64}$$

Now distinguish two cases.

Unbroken generator.

If T_a is unbroken, it leaves the vacuum invariant:

$$T_a \phi_0 = 0. \tag{4.65}$$

Then the equation above is trivial and gives no new massless field.

Broken generator.

If T_a is broken, then it moves the vacuum:

$$T_a \phi_0 \neq 0. \tag{4.66}$$

But the mass-matrix equation says

$$M^2 (T_a \phi_0) = 0. \tag{4.67}$$

Thus $T_a \phi_0$ is a nonzero eigenvector of the mass matrix with eigenvalue zero. A zero eigenvalue of the scalar mass matrix means a massless scalar mode. Therefore each broken generator gives a massless scalar field.

This is the mathematical form of the geometric picture. Broken generators move the vacuum along the manifold of degenerate minima. Motion along that manifold costs no potential energy, so the corresponding direction has zero curvature. Zero curvature of the potential is precisely zero mass squared.

2. The Higgs Mechanism

2.1 Difference between Higgs mechanism and Goldstone's theorem

As we have discussed in last section, Goldstone's theorem should be applied directly only to a *global* continuous symmetry. Suppose a continuous global symmetry is a true physical symmetry of the Lagrangian, but the vacuum does not respect it. Then each broken generator produces one physical massless scalar particle, called a Goldstone boson.

The Higgs mechanism uses a scalar potential with the same geometric shape, but now the symmetry is *local*. A local gauge symmetry is not an ordinary physical symmetry that relates two different states; it is a redundancy in our description of the same physical state. For this reason, Goldstone's theorem does not say that a broken gauge symmetry must produce physical massless particles.

The correct way to think about the Higgs mechanism is the following. First, temporarily ignore the gauge field. The scalar potential then has flat angular directions, and fluctuations along those directions would be Goldstone bosons in the corresponding global theory. After the symmetry is gauged, those same angular fields are no longer physical particles. They can be removed by a gauge choice and become the longitudinal polarizations of the massive gauge bosons. This is why they are called *would-be Goldstone bosons*.

Thus, when we later write a phase field such as $\theta(x)$, we are not claiming that θ is a physical massless Goldstone particle in the gauge theory. We are identifying the angular scalar mode which would have been a Goldstone boson before gauging, and then showing explicitly how the gauge field absorbs it.

In short,

$$\begin{aligned} \text{global spontaneous symmetry breaking} &\Longrightarrow \text{physical massless Goldstone bosons}, \\ \text{gauged version of the same scalar theory} &\Longrightarrow \text{would-be Goldstone modes are eaten}. \end{aligned} \tag{4.68}$$

The result is a massive gauge field. Its extra longitudinal polarization comes from the scalar mode which would have been a Goldstone boson in the ungauged global theory.

2.2 The Abelian Higgs model

We first study the simplest example: a complex scalar field coupled to a $U(1)$ gauge field. The Lagrangian is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + (D_\mu\phi)^*(D^\mu\phi) - V(\phi), \qquad D_\mu = \partial_\mu + \mathrm{i}eA_\mu, \tag{4.69}$$

with

$$V(\phi) = -\mu^2|\phi|^2 + \frac{\lambda}{2}|\phi|^4, \qquad \mu^2 > 0, \quad \lambda > 0. \tag{4.70}$$

Equivalently, the scalar part of the Lagrangian contains

$$\mathcal{L} \supset (D_\mu\phi)^*(D^\mu\phi) + \mu^2|\phi|^2 - \frac{\lambda}{2}|\phi|^4. \tag{4.71}$$

Notice that the sign of quadratic term is positive, which means this is not the mass term yet. The Lagrangian is invariant under the local transformation

$$\phi(x) \longrightarrow e^{\mathrm{i}\alpha(x)}\phi(x), \qquad A_\mu(x) \longrightarrow A_\mu(x) - \frac{1}{e}\partial_\mu\alpha(x). \tag{4.72}$$

The reason for introducing A_μ is exactly to make derivatives compatible with a position-dependent phase. Indeed,

$$D_\mu\phi = (\partial_\mu + \mathrm{i}eA_\mu)\phi \quad \Longrightarrow \quad D_\mu\phi \longrightarrow e^{\mathrm{i}\alpha(x)}D_\mu\phi, \tag{4.73}$$

so $(D_\mu\phi)^*(D^\mu\phi)$ is gauge invariant.

The vacuum

The vacuum is found by minimizing V. Since V depends only on $|\phi|$, write $r = |\phi|$. Then

$$V(r) = -\mu^2r^2 + \frac{\lambda}{2}r^4. \tag{4.74}$$

The stationary condition is

$$\frac{\mathrm{d}V}{\mathrm{d}r} = -2\mu^2r + 2\lambda r^3 = 2r(-\mu^2 + \lambda r^2) = 0. \tag{4.75}$$

Thus either $r = 0$, or

$$r^2 = \frac{\mu^2}{\lambda}. \tag{4.76}$$

The point $r = 0$ is a local maximum, because

$$\left.\frac{\mathrm{d}^2V}{\mathrm{d}r^2}\right|_{r=0} = -2\mu^2 < 0. \tag{4.77}$$

Therefore the minima lie on the circle

$$|\phi| = v, \qquad v^2 = \frac{\mu^2}{\lambda}. \tag{4.78}$$

This circle of minima is the familiar "Mexican-hat" vacuum manifold. The radial direction costs energy, while moving around the circle costs no potential energy before the field is coupled to a gauge field.

Polar form and unitary gauge

Near the vacuum it is useful to separate radial and angular fluctuations:

$$\phi(x) = \left(v + \frac{h(x)}{\sqrt{2}}\right) e^{\mathrm{i}\theta(x)}. \tag{4.79}$$

Here $h(x)$ is the radial fluctuation, while $\theta(x)$ is the angular coordinate along the circle of classical minima. It is important not to misread this sentence: in the local $U(1)$ theory, $\theta(x)$ is not a physical Goldstone particle. Rather, it is the field that *would have been* the Goldstone boson if the same scalar potential had only a global $U(1)$ symmetry. Because the $U(1)$ symmetry is gauged here, $\theta(x)$ is a would-be Goldstone mode and can be removed by a gauge transformation.

To see this explicitly, substitute (4.79) into the covariant derivative:

$$D_\mu\phi = e^{\mathrm{i}\theta}\left[\frac{1}{\sqrt{2}}\partial_\mu h + \mathrm{i}\left(v + \frac{h}{\sqrt{2}}\right)(\partial_\mu\theta + eA_\mu)\right]. \tag{4.80}$$

The phase $e^{\mathrm{i}\theta}$ drops out of the absolute value, so

$$|D_\mu\phi|^2 = \frac{1}{2}(\partial_\mu h)(\partial^\mu h) + e^2\left(v + \frac{h}{\sqrt{2}}\right)^2 B_\mu B^\mu, \qquad B_\mu \equiv A_\mu + \frac{1}{e}\partial_\mu\theta. \tag{4.81}$$

The combination B_μ is what remains after choosing unitary gauge, where the scalar is made real:

$$\phi(x) \longrightarrow v + \frac{h(x)}{\sqrt{2}}, \qquad A_\mu(x) \longrightarrow B_\mu(x). \tag{4.82}$$

Thus the angular field θ has not disappeared mysteriously; it has become part of the gauge field.

Expanding the second term in (4.81) gives

$$e^2\left(v + \frac{h}{\sqrt{2}}\right)^2 B_\mu B^\mu = e^2v^2 B_\mu B^\mu + \sqrt{2}e^2 v\, hB_\mu B^\mu + \frac{e^2}{2}h^2 B_\mu B^\mu. \tag{4.83}$$

The first term is a mass term for the vector field. A massive vector mass term is conventionally written as

$$\frac{1}{2}m_A^2 B_\mu B^\mu. \tag{4.84}$$

Comparing this with $e^2v^2B_\mu B^\mu$, we obtain

$$m_A^2 = 2e^2v^2. \tag{4.85}$$

The remaining terms in (4.83) are interactions between the Higgs fluctuation h and the massive vector boson.

The radial scalar mass follows from the potential. Put

$$r = v + \frac{h}{\sqrt{2}}. \tag{4.86}$$

Since $V'(v) = 0$, the leading nonconstant term is

$$V(r) = V(v) + \frac{1}{2}V''(v)\left(\frac{h}{\sqrt{2}}\right)^2 + \cdots. \tag{4.87}$$

Now

$$V''(r) = -2\mu^2 + 6\lambda r^2, \qquad V''(v) = -2\mu^2 + 6\lambda v^2 = 4\mu^2, \tag{4.88}$$

where $v^2 = \mu^2/\lambda$ has been used. Hence

$$V(r) = V(v) + \mu^2 h^2 + \cdots . \tag{4.89}$$

Because the Lagrangian contains $-V$, the quadratic scalar term is

$$-\mu^2 h^2 = -\frac{1}{2}(2\mu^2)h^2, \tag{4.90}$$

so

$$m_h^2 = 2\mu^2. \tag{4.91}$$

Counting degrees of freedom

Before the Higgs mechanism, the complex scalar has two real degrees of freedom and the massless gauge boson has two transverse polarizations:

$$2 + 2 = 4. \tag{4.92}$$

After choosing the vacuum, only the radial scalar h remains as a physical scalar. The gauge boson is now massive, so it has three polarizations:

$$1 + 3 = 4. \tag{4.93}$$

No physical degree of freedom is lost. The would-be Goldstone mode supplies the longitudinal polarization of the massive vector boson.

	scalar degrees of freedom	vector degrees of freedom	total
before	2	2	4
after	1	3	4

2.3 The same physics in Cartesian fields

The polar form makes the Higgs mechanism transparent, but it is also useful to see how the result appears if we expand the complex scalar into two real fields:

$$\phi(x) = v + \frac{1}{\sqrt{2}}\big(\phi_1(x) + \mathrm{i}\phi_2(x)\big), \qquad \phi_1, \phi_2 \in \mathbb{R}. \tag{4.94}$$

Here ϕ_1 is the radial fluctuation and ϕ_2 is the angular fluctuation to leading order.

To derive the quadratic part of the potential, first compute

$$|\phi|^2 = \left(v + \frac{\phi_1}{\sqrt{2}}\right)^2 + \left(\frac{\phi_2}{\sqrt{2}}\right)^2 = v^2 + \sqrt{2}v\phi_1 + \frac{1}{2}(\phi_1^2 + \phi_2^2). \tag{4.95}$$

It is useful to write this as

$$|\phi|^2 = v^2 + \Delta, \qquad \Delta = \sqrt{2}v\phi_1 + \frac{1}{2}(\phi_1^2 + \phi_2^2). \tag{4.96}$$

Substituting this into $V = -\mu^2|\phi|^2 + \frac{\lambda}{2}|\phi|^4$ gives

$$\begin{aligned} V &= -\mu^2(v^2+\Delta) + \frac{\lambda}{2}(v^2+\Delta)^2 \\ &= V(v) + (-\mu^2 + \lambda v^2)\Delta + \frac{\lambda}{2}\Delta^2. \end{aligned} \tag{4.97}$$

The coefficient of Δ vanishes because the vacuum satisfies $v^2 = \mu^2/\lambda$. Therefore there is no linear term, and the quadratic part comes only from Δ^2. To second order in the fluctuating fields,

$$\Delta^2 = \left(\sqrt{2}v\phi_1\right)^2 + \text{terms of cubic or higher order} = 2v^2\phi_1^2 + \cdots . \tag{4.98}$$

Using $v^2 = \mu^2/\lambda$, the potential therefore expands as

$$V(\phi) = V(v) + \mu^2\phi_1^2 + \text{interaction terms.} \tag{4.99}$$

Therefore

$$-V(\phi) = -V(v) - \mu^2\phi_1^2 + \text{interaction terms.} \tag{4.100}$$

This gives

$$m_{\phi_1}^2 = 2\mu^2, \qquad m_{\phi_2}^2 = 0 \tag{4.101}$$

before gauge fixing. The absence of a ϕ_2^2 term is the usual Goldstone-boson result for the scalar potential. In a gauge theory this field will mix with the gauge field and is not a physical massless particle.

Now expand the covariant derivative:

$$\begin{aligned} D_\mu\phi &= \partial_\mu\phi + \mathrm{i}eA_\mu\phi \\ &= \frac{1}{\sqrt{2}}\left(\partial_\mu\phi_1 + \mathrm{i}\partial_\mu\phi_2\right) + \mathrm{i}evA_\mu + \frac{\mathrm{i}e}{\sqrt{2}}A_\mu(\phi_1 + \mathrm{i}\phi_2). \end{aligned} \tag{4.102}$$

Keeping only terms quadratic in the fields gives

$$|D_\mu\phi|^2 = \frac{1}{2}(\partial_\mu\phi_1)^2 + \frac{1}{2}(\partial_\mu\phi_2)^2 + \sqrt{2}ev\,A^\mu\partial_\mu\phi_2 + e^2v^2A_\mu A^\mu + \cdots . \tag{4.103}$$

The last term is the same gauge-boson mass term as before:

$$e^2v^2A_\mu A^\mu = \frac{1}{2}m_A^2A_\mu A^\mu, \qquad m_A^2 = 2e^2v^2. \tag{4.104}$$

The mixed term

$$\sqrt{2}ev\,A^\mu\partial_\mu\phi_2 \tag{4.105}$$

is the Cartesian-field signal that ϕ_2 is not an independent physical particle. In unitary gauge it is removed. In a covariant R_ξ gauge it is cancelled by a gauge-fixing term, leaving a gauge-dependent unphysical would-be Goldstone field whose effects cancel from physical amplitudes.

2.4 What the propagator is telling us

Now we shall top for a moment and discuss how "the gauge field eats the would-be Goldstone mode" . And what really shows up in the propagator of the gauge field.

Recall the degree-of-freedom count:

$$\text{massless vector: 2 polarizations,} \qquad \text{massive vector: 3 polarizations.} \tag{4.106}$$

The new third polarization is the longitudinal polarization. The question is: where does it come from? The answer is that it comes from the angular scalar mode, namely the would-be Goldstone field.

In unitary gauge the would-be Goldstone field has been removed from the scalar field, so it is hidden inside the gauge field. The massive-vector propagator then takes the form

$$D_{\mu\nu}^{\text{unitary}}(k) = \frac{-\mathrm{i}}{k^2 - m_A^2 + \mathrm{i}\epsilon}\left(g_{\mu\nu} - \frac{k_\mu k_\nu}{m_A^2}\right). \tag{4.107}$$

The term $k_\mu k_\nu/m_A^2$ is the part associated with the longitudinal polarization. Therefore this propagator is the unitary-gauge way of saying: the vector field now contains a longitudinal mode.

The same physics can be seen before choosing unitary gauge. In the Cartesian expansion, the quadratic Lagrangian contains the mixing term

$$\sqrt{2}ev\, A^\mu \partial_\mu \phi_2. \tag{4.108}$$

Using $m_A = \sqrt{2}ev$, this is

$$m_A A^\mu \partial_\mu \phi_2. \tag{4.109}$$

The derivative gives a momentum factor k_μ, so the mixing between A_μ and the would-be Goldstone field carries a factor schematically like $m_A k_\mu$. This is why a would-be Goldstone exchange naturally produces terms proportional to $k_\mu k_\nu$, exactly the tensor structure needed for the longitudinal part of a vector propagator.

Schematically, the gauge-boson mass insertion gives a contribution

$$\mathrm{i}m_A^2 g_{\mu\nu}. \tag{4.110}$$

The exchange of the would-be Goldstone mode gives a longitudinal contribution of the form

$$m_A k_\mu \frac{\mathrm{i}}{k^2}(-m_A k_\nu) = -\mathrm{i}m_A^2 \frac{k_\mu k_\nu}{k^2}. \tag{4.111}$$

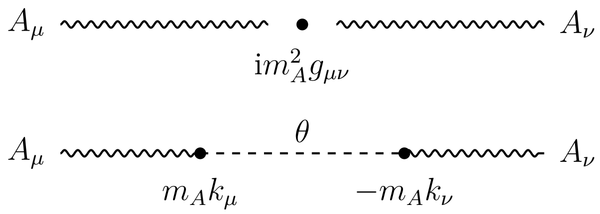


Figure 4.3: The mass insertion and the would-be Goldstone exchange combine to produce the longitudinal structure of the massive gauge boson. This is a schematic diagram; the precise signs depend on the gauge-fixing convention.

Together they form the transverse structure

$$\mathrm{i}m_A^2\left(g_{\mu\nu} - \frac{k_\mu k_\nu}{k^2}\right). \tag{4.112}$$

This equation should be read only as a schematic explanation, not as a complete propagator formula. The lesson is simple: the same scalar mode that disappears in unitary gauge is precisely the mode that supplies the longitudinal polarization of the massive gauge boson. Thus the phrase "the gauge boson eats the Goldstone boson" means:

$$A_\mu \text{ with 2 polarizations} + 1 \text{ would-be Goldstone mode} \implies \text{massive } A_\mu \text{ with 3 polarizations.} \tag{4.113}$$

2.5 $SU(2)$ gauge symmetry breaking

The non-Abelian version uses a complex scalar doublet

$$\Phi = \begin{pmatrix} \Phi_1 \\ \Phi_2 \end{pmatrix}, \qquad \Phi_1, \Phi_2 \in \mathbb{C}, \tag{4.114}$$

coupled to an $SU(2)$ gauge field. The Lagrangian is

$$\mathcal{L} = -\frac{1}{4} G^a_{\mu\nu} G^{a\mu\nu} + (D_\mu \Phi)^\dagger (D^\mu \Phi) - V(\Phi), \tag{4.115}$$

where

$$V(\Phi) = -\mu^2 \Phi^\dagger \Phi + \lambda(\Phi^\dagger \Phi)^2, \qquad \mu^2 > 0, \quad \lambda > 0, \tag{4.116}$$

and

$$D_\mu = \partial_\mu - \mathrm{i} g \frac{\sigma^a}{2} W^a_\mu. \tag{4.117}$$

The σ^a are the Pauli matrices, and repeated adjoint indices $a = 1, 2, 3$ are summed.

The potential depends only on $\Phi^\dagger \Phi$. Let

$$\rho^2 = \Phi^\dagger \Phi. \tag{4.118}$$

Then

$$V(\rho) = -\mu^2 \rho^2 + \lambda \rho^4. \tag{4.119}$$

The minimum satisfies

$$\frac{\mathrm{d}V}{\mathrm{d}\rho} = -2\mu^2 \rho + 4\lambda \rho^3 = 2\rho(-\mu^2 + 2\lambda\rho^2) = 0. \tag{4.120}$$

Thus the nonzero vacuum satisfies

$$\Phi^\dagger \Phi = \rho^2 = \frac{\mu^2}{2\lambda}. \tag{4.121}$$

It is conventional to write

$$\langle \Phi \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}, \qquad v^2 = \frac{\mu^2}{\lambda}. \tag{4.122}$$

Indeed,

$$\langle \Phi \rangle^\dagger \langle \Phi \rangle = \frac{v^2}{2} = \frac{\mu^2}{2\lambda}. \tag{4.123}$$

Notice the dimensions: $\Phi^\dagger \Phi$ has dimension two, so it is proportional to v^2, not to v.

Which generators are broken?

For this pure $SU(2)$ theory, all three generators are broken by the vacuum. Acting with the Pauli matrices gives

$$\sigma^1 \begin{pmatrix} 0 \\ v \end{pmatrix} = \begin{pmatrix} v \\ 0 \end{pmatrix}, \qquad \sigma^2 \begin{pmatrix} 0 \\ v \end{pmatrix} = \begin{pmatrix} -\mathrm{i}v \\ 0 \end{pmatrix}, \qquad \sigma^3 \begin{pmatrix} 0 \\ v \end{pmatrix} = \begin{pmatrix} 0 \\ -v \end{pmatrix}. \tag{4.124}$$

None of these is zero, so the vacuum is moved by every generator. Hence the three would-be Goldstone fields are absorbed by the three $SU(2)$ gauge bosons.

This statement is specific to a pure $SU(2)$ gauge theory with one scalar doublet. In the Standard Model the gauge group is $SU(2)_L \times U(1)_Y$, and one linear combination remains unbroken; that unbroken combination is electromagnetism.

Unitary gauge and the gauge-boson masses

The most general fluctuation around the chosen vacuum may be written as

$$\Phi(x) = \exp\left[\frac{\mathrm{i}}{2}\sigma^a\theta^a(x)\right]\frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v + h(x)\end{pmatrix}. \tag{4.125}$$

The three fields $\theta^a(x)$ are the would-be Goldstone fields. An $SU(2)$ gauge transformation can remove them, leaving unitary gauge:

$$\Phi(x) = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v + h(x)\end{pmatrix}. \tag{4.126}$$

In matrix notation,

$$W_\mu = W_\mu^a\frac{\sigma^a}{2}. \tag{4.127}$$

Under a local $SU(2)$ transformation $U(x)$, the gauge field transforms as

$$W_\mu \longrightarrow UW_\mu U^\dagger - \frac{\mathrm{i}}{g}U\partial_\mu U^\dagger, \tag{4.128}$$

with the sign matching the convention

$$D_\mu = \partial_\mu - \mathrm{i}gW_\mu. \tag{4.129}$$

Now compute the covariant derivative in unitary gauge. Since

$$\sigma^1\begin{pmatrix} 0 \\ v+h\end{pmatrix} = \begin{pmatrix} v+h \\ 0\end{pmatrix}, \quad \sigma^2\begin{pmatrix} 0 \\ v+h\end{pmatrix} = \begin{pmatrix} -\mathrm{i}(v+h) \\ 0\end{pmatrix}, \quad \sigma^3\begin{pmatrix} 0 \\ v+h\end{pmatrix} = \begin{pmatrix} 0 \\ -(v+h)\end{pmatrix}, \tag{4.130}$$

we find

$$D_\mu\Phi = \frac{1}{\sqrt{2}}\begin{pmatrix} -\frac{\mathrm{i}g}{2}(v+h)(W_\mu^1 - \mathrm{i}W_\mu^2) \\ \partial_\mu h + \frac{\mathrm{i}g}{2}(v+h)W_\mu^3\end{pmatrix}. \tag{4.131}$$

Taking the Hermitian norm gives

$$(D_\mu\Phi)^\dagger(D^\mu\Phi) = \frac{1}{2}(\partial_\mu h)(\partial^\mu h) + \frac{g^2}{8}(v+h)^2W_\mu^aW^{a\mu}. \tag{4.132}$$

The gauge-boson mass term is therefore

$$\frac{g^2v^2}{8}W_\mu^aW^{a\mu}. \tag{4.133}$$

Since the canonical mass term for each real vector boson is

$$\frac{1}{2}m_W^2W_\mu^aW^{a\mu}, \tag{4.134}$$

we obtain

$$m_W^2 = \frac{g^2v^2}{4}. \tag{4.135}$$

All three gauge bosons have the same mass because the original theory has an $SU(2)$ symmetry and the scalar representation treats the three generators symmetrically after all of them are broken.

Degree-of-freedom count for the $SU(2)$ model

The complex doublet contains four real scalar degrees of freedom. Before the Higgs mechanism, the three massless gauge bosons each have two transverse polarizations:

$$4 + 3 \times 2 = 10. \tag{4.136}$$

After the Higgs mechanism, one scalar h remains, and the three gauge bosons are massive, each with three polarizations:

$$1 + 3 \times 3 = 10. \tag{4.137}$$

Again the number of physical degrees of freedom is unchanged. The three would-be Goldstone modes have become the longitudinal polarizations of the three massive gauge bosons.

	scalar degrees of freedom	vector degrees of freedom	total
before	4	$3 \times 2 = 6$	10
after	1	$3 \times 3 = 9$	10

3. Chiral Gauge Theories and Fermion Masses

The purpose of this section is to explain why chiral gauge theories are special. The main point is simple:

A chiral gauge symmetry can forbid an ordinary fermion mass term.

This statement is the reason the Higgs mechanism is needed not only for gauge boson masses, but also for fermion masses. A bare Dirac mass would directly couple a left-handed fermion to a right-handed fermion. If those two fields carry different gauge quantum numbers, that coupling is not gauge invariant. The Higgs field supplies exactly the missing gauge quantum numbers. After the Higgs field develops a vacuum expectation value, the Yukawa interaction becomes an ordinary fermion mass term.

3.1 Recap for Dirac spinors and Weyl spinors

A four-component Dirac spinor can be written in terms of two two-component Weyl spinors. In the chiral representation we write

$$\Psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}. \tag{4.138}$$

Here ψ_L is the left-handed Weyl spinor and ψ_R is the right-handed Weyl spinor. The labels L and R refer to chirality. For massless fermions chirality is the same as helicity, but for massive fermions they are not the same concept.

The Dirac adjoint is

$$\bar{\Psi} = \Psi^\dagger \gamma^0. \tag{4.139}$$

With the chiral gamma matrices used below, this becomes

$$\bar{\Psi} = \begin{pmatrix} \psi_R^\dagger & \psi_L^\dagger \end{pmatrix}. \tag{4.140}$$

This reversal of L and R inside $\bar{\Psi}$ is a common source of confusion. It happens because γ^0 is off diagonal in the chiral basis.

We use

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}, \qquad \sigma^\mu = (1, \vec{\sigma}), \qquad \bar{\sigma}^\mu = (1, -\vec{\sigma}), \tag{4.141}$$

where $\vec{\sigma}$ are the Pauli matrices. In this convention

$$\gamma^5 = \mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{4.142}$$

Therefore the chiral projection operators are

$$P_L = \frac{1-\gamma^5}{2}, \qquad P_R = \frac{1+\gamma^5}{2}. \tag{4.143}$$

They obey

$$P_L^2 = P_L, \qquad P_R^2 = P_R, \qquad P_L P_R = 0, \qquad P_L + P_R = 1. \tag{4.144}$$

Thus

$$\Psi_L = P_L \Psi, \qquad \Psi_R = P_R \Psi. \tag{4.145}$$

Some books define γ^5 with the opposite sign. Then the displayed formulas for P_L and P_R are interchanged. This is only a convention; one must simply use it consistently.

Now we can write the free Dirac Lagrangian is

$$\mathcal{L} = \bar{\Psi}\left(\mathrm{i}\gamma^\mu \partial_\mu - m\right)\Psi. \tag{4.146}$$

Using the block form of the gamma matrices, the kinetic term splits into a left-handed part and a right-handed part:

$$\begin{aligned} \bar{\Psi}\,\mathrm{i}\gamma^\mu\partial_\mu\Psi &= \begin{pmatrix} \psi_R^\dagger & \psi_L^\dagger \end{pmatrix} \mathrm{i} \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix} \partial_\mu \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} \\ &= \mathrm{i}\psi_R^\dagger \sigma^\mu \partial_\mu \psi_R + \mathrm{i}\psi_L^\dagger \bar{\sigma}^\mu \partial_\mu \psi_L. \end{aligned} \tag{4.147}$$

The mass term, however, mixes left and right:

$$\begin{aligned} -m\bar{\Psi}\Psi &= -m \begin{pmatrix} \psi_R^\dagger & \psi_L^\dagger \end{pmatrix} \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} \\ &= -m\left(\psi_R^\dagger \psi_L + \psi_L^\dagger \psi_R\right). \end{aligned} \tag{4.148}$$

Equivalently, in four-component notation,

$$-m\bar{\Psi}\Psi = -m\left(\bar{\Psi}_L \Psi_R + \bar{\Psi}_R \Psi_L\right). \tag{4.149}$$

This is the key algebraic fact. A fermion mass term is not a purely left-handed or purely right-handed object. It couples the two chiralities.

3.2 Chiral gauge theory

A gauge theory is called *chiral* when the left-handed and right-handed fermions transform differently under the gauge group. The simplest example is a $U(1)$ gauge symmetry. Let

$$\psi_L(x) \longrightarrow \psi_L'(x) = e^{\mathrm{i}y_L\alpha(x)}\psi_L(x), \qquad \psi_R(x) \longrightarrow \psi_R'(x) = e^{\mathrm{i}y_R\alpha(x)}\psi_R(x). \tag{4.150}$$

The function $\alpha(x)$ is the same local gauge parameter, but the charges y_L and y_R may be different. If

$$y_L = y_R, \tag{4.151}$$

then the theory is vector-like. Ordinary QED is of this type: the left- and right-handed parts of the electron carry the same electric charge. If

$$y_L \neq y_R, \tag{4.152}$$

then the theory is chiral.

With the convention

$$\psi(x) \longrightarrow e^{\mathrm{i}y\alpha(x)}\psi(x), \tag{4.153}$$

the covariant derivative is

$$D_\mu\psi = (\partial_\mu - \mathrm{i}gyA_\mu)\,\psi, \tag{4.154}$$

provided the gauge field transforms as

$$A_\mu(x) \longrightarrow A_\mu'(x) = A_\mu(x) + \frac{1}{g}\partial_\mu\alpha(x). \tag{4.155}$$

Indeed,

$$\begin{aligned}
D_\mu'\psi' &= \left(\partial_\mu - \mathrm{i}gyA_\mu'\right) e^{\mathrm{i}y\alpha}\psi \\
&= e^{\mathrm{i}y\alpha}\left[\partial_\mu + \mathrm{i}y\partial_\mu\alpha - \mathrm{i}gy\left(A_\mu + \frac{1}{g}\partial_\mu\alpha\right)\right]\psi \\
&= e^{\mathrm{i}y\alpha}\left(\partial_\mu - \mathrm{i}gyA_\mu\right)\psi \\
&= e^{\mathrm{i}y\alpha}D_\mu\psi.
\end{aligned} \tag{4.156}$$

Thus $D_\mu\psi$ transforms in the same way as ψ, which is precisely why the kinetic term is gauge invariant.

Now examine the mass term in a chiral $U(1)$ theory. Since

$$\bar\psi_L \longrightarrow e^{-\mathrm{i}y_L\alpha(x)}\bar\psi_L, \qquad \psi_R \longrightarrow e^{\mathrm{i}y_R\alpha(x)}\psi_R, \tag{4.157}$$

the bilinear transforms as

$$\bar\psi_L\psi_R \longrightarrow e^{\mathrm{i}(y_R-y_L)\alpha(x)}\bar\psi_L\psi_R. \tag{4.158}$$

Similarly,

$$\bar\psi_R\psi_L \longrightarrow e^{\mathrm{i}(y_L-y_R)\alpha(x)}\bar\psi_R\psi_L. \tag{4.159}$$

Therefore

$$m\left(\bar\psi_L\psi_R + \bar\psi_R\psi_L\right) \tag{4.160}$$

is gauge invariant only if

$$y_L = y_R. \tag{4.161}$$

If $y_L \neq y_R$, an explicit Dirac mass is forbidden by gauge invariance. This is what the short phrase "a chiral gauge theory requires $m = 0$" really means: the Lagrangian cannot contain a bare Dirac mass before symmetry breaking. It does *not* mean that the physical fermion must remain massless forever.

3.3 The Higgs-Yukawa solution in a $U(1)$ toy model

The Higgs field can repair the gauge transformation of the forbidden mass term. Introduce a complex scalar field ϕ and write the Yukawa interaction

$$\mathcal{L}_{\text{Yuk}} = -\lambda\left(\phi\,\bar{\psi}_L\psi_R + \phi^*\,\bar{\psi}_R\psi_L\right). \tag{4.162}$$

For this to be gauge invariant, ϕ must transform as

$$\phi \longrightarrow \phi' = e^{\mathrm{i}(y_L-y_R)\alpha(x)}\phi. \tag{4.163}$$

Then

$$\begin{aligned}\phi\,\bar{\psi}_L\psi_R &\longrightarrow e^{\mathrm{i}(y_L-y_R)\alpha}\phi\, e^{\mathrm{i}(y_R-y_L)\alpha}\bar{\psi}_L\psi_R\\ &= \phi\,\bar{\psi}_L\psi_R.\end{aligned} \tag{4.164}$$

The Higgs field carries exactly the difference between the left-handed and right-handed charges.

Choose the usual symmetry-breaking potential

$$V(\phi) = -\mu^2\phi^*\phi + \frac{\lambda_\phi}{2}\left(\phi^*\phi\right)^2, \qquad \mu^2>0, \qquad \lambda_\phi>0. \tag{4.165}$$

The minimum satisfies

$$\begin{aligned}\frac{\partial V}{\partial(\phi^*\phi)} &= -\mu^2 + \lambda_\phi(\phi^*\phi) = 0,\\ \langle\phi^*\phi\rangle &= \frac{\mu^2}{\lambda_\phi}.\end{aligned} \tag{4.166}$$

Writing

$$\phi(x) = \frac{1}{\sqrt{2}}\left(v + h(x)\right)e^{\mathrm{i}\beta(x)/v}, \tag{4.167}$$

we have

$$\langle\phi^*\phi\rangle = \frac{v^2}{2}. \tag{4.168}$$

Hence

$$\frac{v^2}{2} = \frac{\mu^2}{\lambda_\phi}, \qquad v^2 = \frac{2\mu^2}{\lambda_\phi}. \tag{4.169}$$

In unitary gauge the phase $\beta(x)$ is removed, and the Yukawa term becomes

$$\begin{aligned}\mathcal{L}_{\text{Yuk}} &= -\lambda\frac{v+h(x)}{\sqrt{2}}\left(\bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L\right)\\ &= -\frac{\lambda v}{\sqrt{2}}\bar{\psi}\psi - \frac{\lambda}{\sqrt{2}}h\bar{\psi}\psi.\end{aligned} \tag{4.170}$$

Therefore the fermion mass is

$$m_\psi = \frac{\lambda v}{\sqrt{2}}. \tag{4.171}$$

The same interaction also predicts a Higgs-fermion coupling:

$$\mathcal{L}_{h\bar{\psi}\psi} = -\frac{m_\psi}{v}h\bar{\psi}\psi. \tag{4.172}$$

This relation is conceptually important. In a Higgs theory, fermion masses and Higgs-fermion couplings have the same origin.

3.4 Non-Abelian chiral gauge theories

The same idea works for a non-Abelian gauge group. Let the left-handed and right-handed fermions transform in possibly different representations:

$$\psi_L \longrightarrow U_L(x)\psi_L, \qquad \psi_R \longrightarrow U_R(x)\psi_R. \tag{4.173}$$

For infinitesimal transformations,

$$U_L(x) = 1 + \mathrm{i}\alpha^a(x)t_L^a, \qquad U_R(x) = 1 + \mathrm{i}\alpha^a(x)t_R^a, \tag{4.174}$$

so

$$\delta\psi_L = \mathrm{i}\alpha^a t_L^a \psi_L, \qquad \delta\psi_R = \mathrm{i}\alpha^a t_R^a \psi_R. \tag{4.175}$$

The covariant derivatives are

$$D_\mu\psi_L = \left(\partial_\mu - \mathrm{i}gA_\mu^a t_L^a\right)\psi_L, \qquad D_\mu\psi_R = \left(\partial_\mu - \mathrm{i}gA_\mu^a t_R^a\right)\psi_R. \tag{4.176}$$

Here t_L^a and t_R^a are matrices appropriate to the representations of ψ_L and ψ_R. They need not be the same matrices.

A bare mass term transforms schematically as

$$\bar{\psi}_L\psi_R \longrightarrow \bar{\psi}_L U_L^\dagger U_R \psi_R. \tag{4.177}$$

For this to be invariant for every gauge transformation, one needs

$$U_L^\dagger U_R = 1 \tag{4.178}$$

on the relevant fermion indices. This happens when the two chiralities are in equivalent representations. If they are in different representations, the bare mass term is not a gauge singlet and is forbidden.

The Higgs field must be chosen so that the Yukawa product is a gauge singlet:

$$\mathcal{L}_{\text{Yuk}} = -y\left(\bar{\psi}_L\,\Phi\,\psi_R + \text{h.c.}\right). \tag{4.179}$$

In representation language, this means that the product

$$R_L^* \otimes R_\Phi \otimes R_R \tag{4.180}$$

must contain the singlet representation. This is the non-Abelian version of the statement that the Higgs field carries the missing gauge charge.

There is one further consistency condition that is often not written in short notes: a quantum chiral gauge theory must also be anomaly free. Anomaly cancellation is separate from the classical gauge invariance discussed here. The Standard Model has precisely arranged fermion quantum numbers so that the gauge anomalies cancel.

3.5 The Glashow-Salam-Weinberg model

The electroweak theory is the most important example. Its gauge group is

$$G_{\text{EW}} = SU(2)_L \times U(1)_Y. \tag{4.181}$$

The Higgs mechanism breaks it to electromagnetism:

$$SU(2)_L \times U(1)_Y \longrightarrow U(1)_{\text{em}}. \tag{4.182}$$

The unbroken electric charge generator is

$$Q = T^3 + Y, \tag{4.183}$$

where $T^3 = \sigma^3/2$ for an $SU(2)$ doublet and Y is hypercharge.

One lepton family

For one lepton family, the field content is

field	$SU(2)_L$ representation	Y	electric charge Q
W^a_μ, $a = 1, 2, 3$	$\mathbf{3}$	0	$0, \pm 1$ after $W^\pm$
B_μ	$\mathbf{1}$	0	0
$L = \begin{pmatrix} \nu_e \\ e \end{pmatrix}_L$	$\mathbf{2}$	$-\frac{1}{2}$	$\begin{pmatrix} 0 \\ -1 \end{pmatrix}$
e_R	$\mathbf{1}$	-1	-1
$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}$	$\mathbf{2}$	$\frac{1}{2}$	$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

The charges in the last column follow from $Q = T^3 + Y$. For example, for the lepton doublet

$$Q_L = \frac{\sigma^3}{2} - \frac{1}{2}\mathbf{1} = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}. \tag{4.184}$$

For the Higgs doublet,

$$Q_\Phi = \frac{\sigma^3}{2} + \frac{1}{2}\mathbf{1} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \tag{4.185}$$

Thus the lower component ϕ^0 is electrically neutral. This is why it is allowed to develop a vacuum expectation value without breaking $U(1)_{\text{em}}$.

The electroweak Lagrangian

For the lepton-Higgs sector, the Lagrangian is

$$\begin{aligned} \mathcal{L} = &-\frac{1}{4}W^a_{\mu\nu}W^{a\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu} + (D_\mu\Phi)^\dagger(D^\mu\Phi) - V(\Phi) \\ &+ \mathrm{i}\bar{L}\bar{\sigma}^\mu D_\mu L + \mathrm{i}\bar{e}_R\sigma^\mu D_\mu e_R - y_e\left(\bar{L}\Phi e_R + \bar{e}_R\Phi^\dagger L\right). \end{aligned} \tag{4.186}$$

The covariant derivative acting on a field with weak isospin generators T^a and hypercharge Y is

$$D_\mu = \partial_\mu - \mathrm{i}gW^a_\mu T^a - \mathrm{i}g'YB_\mu. \tag{4.187}$$

For an $SU(2)$ doublet,

$$T^a = \frac{\sigma^a}{2}. \tag{4.188}$$

For a singlet, $T^a = 0$.

The electron Yukawa term is gauge invariant for two independent reasons. First, the $SU(2)$ indices can be contracted:

$$\bar{L}\Phi = \bar{L}_i \Phi^i. \tag{4.189}$$

Second, the hypercharges add to zero:

$$Y(\bar{L}) + Y(\Phi) + Y(e_R) = \frac{1}{2} + \frac{1}{2} - 1 = 0. \tag{4.190}$$

By contrast, the bare mass term $\bar{L}e_R$ is not gauge invariant:

$$\bar{L}e_R \quad \text{is an } SU(2)_L \text{ doublet, not a singlet.} \tag{4.191}$$

This is the electroweak version of the earlier $U(1)$ lesson.

Gauge boson masses

The Higgs potential is

$$V(\Phi) = -\mu^2 \Phi^\dagger \Phi + \lambda \left(\Phi^\dagger \Phi\right)^2. \tag{4.192}$$

The vacuum can be chosen as

$$\langle \Phi \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}. \tag{4.193}$$

Since the lower component has $Q = 0$, this vacuum preserves electromagnetism.

Insert the vacuum into the Higgs kinetic term:

$$\begin{aligned} D_\mu \langle \Phi \rangle &= \left(-\mathrm{i}gW_\mu^a \frac{\sigma^a}{2} - \mathrm{i}g' \frac{1}{2} B_\mu\right) \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix} \\ &= -\frac{\mathrm{i}v}{2\sqrt{2}} \begin{pmatrix} g(W_\mu^1 - \mathrm{i}W_\mu^2) \\ -gW_\mu^3 + g'B_\mu \end{pmatrix}. \end{aligned} \tag{4.194}$$

Therefore

$$(D_\mu \langle \Phi \rangle)^\dagger (D^\mu \langle \Phi \rangle) = \frac{g^2 v^2}{8} \left[(W_\mu^1)^2 + (W_\mu^2)^2\right] + \frac{v^2}{8} \left(gW_\mu^3 - g'B_\mu\right)^2. \tag{4.195}$$

Define

$$W_\mu^\pm = \frac{1}{\sqrt{2}} \left(W_\mu^1 \mp \mathrm{i}W_\mu^2\right). \tag{4.196}$$

Then

$$\frac{g^2 v^2}{8} \left[(W_\mu^1)^2 + (W_\mu^2)^2\right] = \frac{g^2 v^2}{4} W_\mu^+ W^{-\mu}. \tag{4.197}$$

For a charged complex vector field the mass term has the form

$$m_W^2 W_\mu^+ W^{-\mu}. \tag{4.198}$$

Thus

$$m_W = \frac{gv}{2}. \tag{4.199}$$

For the neutral gauge bosons, define the weak mixing angle by

$$\tan \theta_W = \frac{g'}{g}. \tag{4.200}$$

The massive and massless neutral fields are

$$Z_\mu = \cos\theta_W\, W^3_\mu - \sin\theta_W\, B_\mu, \qquad A_\mu = \sin\theta_W\, W^3_\mu + \cos\theta_W\, B_\mu. \tag{4.201}$$

The mass term becomes

$$\frac{v^2}{8}\left(gW^3_\mu - g' B_\mu\right)^2 = \frac{v^2}{8}(g^2 + g'^2) Z_\mu Z^\mu. \tag{4.202}$$

For a real vector field the mass term has the form

$$\frac{1}{2} m_Z^2 Z_\mu Z^\mu. \tag{4.203}$$

Hence

$$m_Z = \frac{v}{2}\sqrt{g^2 + g'^2}, \qquad m_A = 0. \tag{4.204}$$

The photon remains massless because it is the gauge boson of the unbroken $U(1)_{\text{em}}$.

The weak mixing angle also satisfies

$$\cos\theta_W = \frac{g}{\sqrt{g^2+g'^2}} = \frac{m_W}{m_Z}, \qquad \sin\theta_W = \frac{g'}{\sqrt{g^2+g'^2}}. \tag{4.205}$$

The electric charge is

$$e = g\sin\theta_W = g'\cos\theta_W = \frac{gg'}{\sqrt{g^2+g'^2}}. \tag{4.206}$$

Covariant derivatives after mixing

It is useful to rewrite the neutral part of the covariant derivative in terms of A_μ and Z_μ. Starting from

$$D_\mu = \partial_\mu - \mathrm{i} g W^a_\mu T^a - \mathrm{i} g' Y B_\mu, \tag{4.207}$$

use

$$W^3_\mu = \cos\theta_W Z_\mu + \sin\theta_W A_\mu, \qquad B_\mu = -\sin\theta_W Z_\mu + \cos\theta_W A_\mu. \tag{4.208}$$

The A_μ coefficient is

$$\begin{aligned} -\mathrm{i}\left(g\sin\theta_W T^3 + g'\cos\theta_W Y\right) A_\mu &= -\mathrm{i}e(T^3 + Y)A_\mu \\ &= -\mathrm{i}eQA_\mu. \end{aligned} \tag{4.209}$$

This proves that the massless field A_μ couples to electric charge.

The Z_μ coefficient is

$$-\mathrm{i}\left(g\cos\theta_W T^3 - g'\sin\theta_W Y\right) Z_\mu = -\mathrm{i}\frac{g}{\cos\theta_W}\left(T^3 - \sin^2\theta_W Q\right) Z_\mu. \tag{4.210}$$

Thus, for a weak doublet,

$$\begin{aligned} D_\mu L = \Bigg[&\partial_\mu - \mathrm{i}\frac{g}{\sqrt{2}}\left(W^+_\mu \sigma^+ + W^-_\mu \sigma^-\right) \\ &- \mathrm{i}\frac{g}{\cos\theta_W} Z_\mu\left(T^3 - \sin^2\theta_W Q\right) - \mathrm{i}eA_\mu Q\Bigg] L, \end{aligned} \tag{4.211}$$

where

$$\sigma^\pm = \frac{1}{2}\left(\sigma^1 \pm \mathrm{i}\sigma^2\right). \tag{4.212}$$

For the right-handed electron, $T^3 = 0$ and $Q = -1$, so

$$D_\mu e_R = \left[\partial_\mu - \mathrm{i}\frac{g}{\cos\theta_W}\sin^2\theta_W Z_\mu + \mathrm{i}eA_\mu\right] e_R. \tag{4.213}$$

Electron mass from the Higgs field

In unitary gauge,

$$\Phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}. \tag{4.214}$$

Then the electron Yukawa term becomes

$$\begin{aligned} -y_e \left(\bar{L} \Phi e_R + \bar{e}_R \Phi^\dagger L \right) &= -y_e \frac{v+h}{\sqrt{2}} \left(\bar{e}_L e_R + \bar{e}_R e_L \right) \\ &= -m_e \bar{e} e - \frac{m_e}{v} h \bar{e} e. \end{aligned} \tag{4.215}$$

Therefore

$$m_e = \frac{y_e v}{\sqrt{2}}. \tag{4.216}$$

The neutrino remains massless in this minimal lepton model because there is no right-handed neutrino field and therefore no analogous renormalizable Dirac Yukawa term.

3.6 Summary of the logic

The whole construction can be compressed into the following chain:

- different gauge quantum numbers for $\psi_L, \psi_R \Longrightarrow$no gauge-invariant bare Dirac mass,
- Higgs field carries the missing quantum numbers $\Longrightarrow$ gauge-invariant Yukawa interaction,
- $\langle \Phi \rangle \neq 0 \Longrightarrow$ fermion mass after symmetry breaking.

This is why the Higgs mechanism is tied so deeply to chiral gauge theories. It does not merely give masses to $W^\pm$ and Z; it also allows fermion masses without explicitly violating the gauge symmetry.

4. The Standard Model

The Standard Model is the quantum field theory of the strong, weak, and electromagnetic interactions. Its starting point is the gauge group

$$G_{\text{SM}} = SU(3)_c \times SU(2)_L \times U(1)_Y. \tag{4.217}$$

Each factor has a distinct physical meaning. The group $SU(3)_c$ is the color gauge group of QCD. The group $SU(2)_L$ acts on left-handed weak doublets. The group $U(1)_Y$ is the hypercharge gauge group. The subscript L in $SU(2)_L$ is not decorative: it tells us that the weak interaction distinguishes left-handed and right-handed fermions.

The Higgs field breaks only the electroweak part of the gauge group:

$$SU(2)_L \times U(1)_Y \longrightarrow U(1)_{\text{em}}. \tag{4.218}$$

Color is not broken, so after electroweak symmetry breaking the unbroken gauge symmetry is

$$SU(3)_c \times U(1)_{\text{em}}. \tag{4.219}$$

The generator of the remaining electromagnetic gauge group is

$$Q = T^3 + Y. \tag{4.220}$$

Here T^3 is the third weak-isospin generator and Y is hypercharge. For an $SU(2)_L$ doublet,

$$T^3 = \frac{\sigma^3}{2} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}. \tag{4.221}$$

For an $SU(2)_L$ singlet, $T^3 = 0$, so its electric charge is simply equal to its hypercharge.

4.1 Field content

Representations will be denoted by their dimensions:

$$\mathbf{1}, \qquad \mathbf{2}, \qquad \mathbf{3}, \qquad \mathbf{8}. \tag{4.222}$$

For $SU(3)_c$, $\mathbf{3}$ is the fundamental color representation, $\overline{\mathbf{3}}$ is the anti-fundamental representation, and $\mathbf{8}$ is the adjoint representation. For $SU(2)_L$, $\mathbf{2}$ is a weak doublet, $\mathbf{1}$ is a weak singlet, and $\mathbf{3}$ is the adjoint representation.

The gauge bosons are fixed by the gauge group. The gluons G^a_μ are the gauge bosons of $SU(3)_c$, the fields W^a_μ are the gauge bosons of $SU(2)_L$, and B_μ is the gauge boson of $U(1)_Y$:

field	$SU(3)_c$	$SU(2)_L$	Y	electric charge
$G^a_\mu,\ a = 1,\ldots,8$	$\mathbf{8}$	$\mathbf{1}$	0	0
$W^a_\mu,\ a = 1,2,3$	$\mathbf{1}$	$\mathbf{3}$	0	$0, \pm 1$ after mixing
B_μ	$\mathbf{1}$	$\mathbf{1}$	0	0

The charged weak bosons are not W^1_μ and W^2_μ separately. They are the complex combinations

$$W^\pm_\mu = \frac{1}{\sqrt{2}}\left(W^1_\mu \mp \mathrm{i} W^2_\mu\right). \tag{4.223}$$

The remaining neutral fields W^3_μ and B_μ later combine into the photon and the Z boson.

There are three families of fermions:

$$I = 1, 2, 3. \tag{4.224}$$

For leptons these are the e, μ, τ families. For quarks they are

$$(u, d), \qquad (c, s), \qquad (t, b). \tag{4.225}$$

In each family the left-handed lepton fields form a weak doublet,

$$L^I_L = \begin{pmatrix} \nu^I_L \\ e^I_L \end{pmatrix}, \qquad L^I_L \sim (\mathbf{1}, \mathbf{2})_{-\frac{1}{2}}, \tag{4.226}$$

while the right-handed charged lepton is a weak singlet,

$$e^I_R \sim (\mathbf{1}, \mathbf{1})_{-1}. \tag{4.227}$$

The electric charges follow from $Q = T^3 + Y$. For the lepton doublet,

$$Q(L_L) = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}. \tag{4.228}$$

Thus $Q(\nu_L) = 0$ and $Q(e_L) = -1$. Since e_R is a weak singlet,

$$Q(e_R) = Y(e_R) = -1. \tag{4.229}$$

For quarks, the left-handed fields form a color triplet and weak doublet:

$$Q_L^I = \begin{pmatrix} u_L^I \\ d_L^I \end{pmatrix}, \qquad Q_L^I \sim (\mathbf{3}, \mathbf{2})_{\frac{1}{6}}. \tag{4.230}$$

The right-handed quarks are weak singlets:

$$u_R^I \sim (\mathbf{3}, \mathbf{1})_{\frac{2}{3}}, \qquad d_R^I \sim (\mathbf{3}, \mathbf{1})_{-\frac{1}{3}}. \tag{4.231}$$

For the left-handed quark doublet,

$$Q(Q_L) = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} + \frac{1}{6}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{2}{3} & 0 \\ 0 & -\frac{1}{3} \end{pmatrix}. \tag{4.232}$$

Therefore u-type quarks have electric charge $2/3$, and d-type quarks have electric charge $-1/3$.

The scalar sector contains one complex Higgs doublet,

$$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}, \qquad \Phi \sim (\mathbf{1}, \mathbf{2})_{\frac{1}{2}}. \tag{4.233}$$

Again using $Q = T^3 + Y$,

$$Q_\Phi = \frac{\sigma^3}{2} + \frac{1}{2}\mathbf{1} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \tag{4.234}$$

The lower component ϕ^0 is electrically neutral. This is essential: only a neutral field can acquire a vacuum expectation value without breaking electromagnetism.

sector	field	$SU(3)_c$	$SU(2)_L$	Y	Q
gauge bosons	$G_\mu^a,\ a = 1, \ldots, 8$	$\mathbf{8}$	$\mathbf{1}$	0	0
	$W_\mu^a,\ a = 1, 2, 3$	$\mathbf{1}$	$\mathbf{3}$	0	$W^\pm : \pm 1,\ W^3 : 0$
	B_μ	$\mathbf{1}$	$\mathbf{1}$	0	0
leptons	$L_L^I = (\nu_L^I, e_L^I)^T$	$\mathbf{1}$	$\mathbf{2}$	$-\frac{1}{2}$	$(0, -1)$
	e_R^I	$\mathbf{1}$	$\mathbf{1}$	-1	-1
quarks	$Q_L^I = (u_L^I, d_L^I)^T$	$\mathbf{3}$	$\mathbf{2}$	$\frac{1}{6}$	$(\frac{2}{3}, -\frac{1}{3})$
	u_R^I	$\mathbf{3}$	$\mathbf{1}$	$\frac{2}{3}$	$\frac{2}{3}$
	d_R^I	$\mathbf{3}$	$\mathbf{1}$	$-\frac{1}{3}$	$-\frac{1}{3}$
Higgs	$\Phi = (\phi^+, \phi^0)^T$	$\mathbf{1}$	$\mathbf{2}$	$\frac{1}{2}$	$(1, 0)$

表 4.1: Minimal Standard Model field content. The family index is $I = 1, 2, 3$. The electric charges in the last column are obtained from $Q = T^3 + Y$.

4.2 The Lagrangian before symmetry breaking

The Standard Model Lagrangian is built from gauge kinetic terms, fermion kinetic terms, the Higgs kinetic and potential terms, and Yukawa interactions:

$$\mathcal{L}_{\text{SM}} = \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{fermion}} + \mathcal{L}_{\text{Higgs}} + \mathcal{L}_{\text{Yukawa}}. \tag{4.235}$$

The gauge kinetic terms are

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4}G^a_{\mu\nu}G^{a\mu\nu} - \frac{1}{4}W^a_{\mu\nu}W^{a\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu}. \tag{4.236}$$

The indices are $a = 1, \dots, 8$ for $SU(3)_c$ and $a = 1, 2, 3$ for $SU(2)_L$.

The covariant derivative tells us how every field couples to the gauge bosons:

$$D_\mu = \partial_\mu - \mathrm{i}g_s G^a_\mu T^a_c - \mathrm{i}g W^a_\mu T^a_L - \mathrm{i}g' Y B_\mu. \tag{4.237}$$

If a field is a singlet under a group, the corresponding generator is zero. For example, e_R is a singlet under $SU(3)_c$ and $SU(2)_L$, so only the hypercharge term acts on e_R.

In four-component notation the fermion kinetic terms are

$$\begin{aligned}\mathcal{L}_{\text{fermion}} &= \mathrm{i}\sum_{I=1}^{3}\left[\bar{L}^I_L\gamma^\mu D_\mu L^I_L + \bar{e}^I_R\gamma^\mu D_\mu e^I_R\right] \\ &\quad + \mathrm{i}\sum_{I=1}^{3}\left[\bar{Q}^I_L\gamma^\mu D_\mu Q^I_L + \bar{u}^I_R\gamma^\mu D_\mu u^I_R + \bar{d}^I_R\gamma^\mu D_\mu d^I_R\right].\end{aligned} \tag{4.238}$$

This form is compact, but the left-handed and right-handed labels should not be ignored: the two chiralities carry different electroweak quantum numbers.

The Higgs part is

$$\mathcal{L}_{\text{Higgs}} = (D_\mu\Phi)^\dagger(D^\mu\Phi) - V(\Phi), \tag{4.239}$$

with

$$V(\Phi) = -\mu^2\Phi^\dagger\Phi + \lambda\left(\Phi^\dagger\Phi\right)^2. \tag{4.240}$$

The Yukawa interactions are the gauge-invariant substitutes for ordinary fermion mass terms. The Standard Model does not allow a bare term such as $\bar{L}_L e_R$, because L_L is a weak doublet and e_R is a weak singlet. The Higgs doublet supplies the missing weak-isospin and hypercharge quantum numbers.

For up-type masses one needs the conjugate Higgs doublet

$$\widetilde{\Phi} = \mathrm{i}\sigma^2\Phi^* = \begin{pmatrix}\phi^{0*}\\ -\phi^-\end{pmatrix}, \qquad \widetilde{\Phi} \sim (\mathbf{1}, \mathbf{2})_{-\frac{1}{2}}. \tag{4.241}$$

The full Yukawa interaction is

$$\begin{aligned}\mathcal{L}_{\text{Yukawa}} &= -(Y_e)_{IJ}\bar{L}^I_L\Phi e^J_R - (Y_d)_{IJ}\bar{Q}^I_L\Phi d^J_R \\ &\quad - (Y_u)_{IJ}\bar{Q}^I_L\widetilde{\Phi}u^J_R + \text{h.c.}\end{aligned} \tag{4.242}$$

The matrices Y_e, Y_d, Y_u act in family space.

The hypercharge checks show why these terms have exactly this form. Since $Y(\bar{\psi}) = -Y(\psi)$, the electron Yukawa term has

$$Y(\bar{L}_L) + Y(\Phi) + Y(e_R) = \frac{1}{2} + \frac{1}{2} - 1 = 0. \tag{4.243}$$

The down-quark Yukawa term has

$$Y(\bar{Q}_L) + Y(\Phi) + Y(d_R) = -\frac{1}{6} + \frac{1}{2} - \frac{1}{3} = 0, \tag{4.244}$$

and the up-quark Yukawa term has

$$Y(\bar{Q}_L) + Y(\widetilde{\Phi}) + Y(u_R) = -\frac{1}{6} - \frac{1}{2} + \frac{2}{3} = 0. \tag{4.245}$$

Thus every Yukawa term is a gauge singlet.

4.3 Electroweak symmetry breaking and currents

The Higgs vacuum can be chosen as

$$\langle\Phi\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\ v\end{pmatrix}. \tag{4.246}$$

In unitary gauge,

$$\Phi(x) = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\ v+h(x)\end{pmatrix}, \qquad \widetilde{\Phi}(x) = \frac{1}{\sqrt{2}}\begin{pmatrix}v+h(x)\\ 0\end{pmatrix}. \tag{4.247}$$

The neutral gauge bosons mix according to

$$A_\mu = \sin\theta_W W^3_\mu + \cos\theta_W B_\mu, \qquad Z_\mu = \cos\theta_W W^3_\mu - \sin\theta_W B_\mu, \tag{4.248}$$

where

$$\tan\theta_W = \frac{g'}{g}, \qquad e = g\sin\theta_W = g'\cos\theta_W. \tag{4.249}$$

The field A_μ is the photon and remains massless; Z_μ is massive.

For a weak doublet, the broken-phase covariant derivative can be written as

$$\begin{aligned} D_\mu = \bar{D}_\mu &- \mathrm{i}\frac{g}{\sqrt{2}}\left(W^+_\mu T^+ + W^-_\mu T^-\right) \\ &- \mathrm{i}\frac{g}{\cos\theta_W} Z_\mu\left(T^3 - \sin^2\theta_W Q\right) - \mathrm{i}eA_\mu Q. \end{aligned} \tag{4.250}$$

Here $\bar{D}_\mu$ contains the color part when the field is a quark, and

$$T^\pm = \frac{\sigma^1 \pm \mathrm{i}\sigma^2}{2}. \tag{4.251}$$

The $W^\pm$ terms change the upper component of a weak doublet into the lower component or vice versa. The photon term couples to electric charge. The Z term couples to the combination $T^3 - \sin^2\theta_W Q$.

After symmetry breaking, the fermion-gauge interactions can be summarized by

$$\begin{aligned} \mathcal{L}_{\text{int}} = \frac{g}{\sqrt{2}}&\left(W^+_\mu J^\mu_+ + W^-_\mu J^\mu_-\right) + \frac{g}{\cos\theta_W} Z_\mu J^\mu_Z \\ &+ eA_\mu J^\mu_{\text{em}} + g_s G^a_\mu J^\mu_a. \end{aligned} \tag{4.252}$$

Ignoring quark mixing, the charged currents are

$$J^\mu_+ = \sum_{I=1}^{3}\left(\bar{\nu}^I_L\gamma^\mu e^I_L + \bar{u}^I_L\gamma^\mu d^I_L\right), \tag{4.253}$$

and

$$J^\mu_- = \sum_{I=1}^{3}\left(\bar{e}^I_L\gamma^\mu \nu^I_L + \bar{d}^I_L\gamma^\mu u^I_L\right). \tag{4.254}$$

The charged current is purely left-handed. In the quark sector, the realistic charged current contains the CKM matrix:

$$\bar{u}^I_L\gamma^\mu d^I_L \quad\longrightarrow\quad \bar{u}^I_L\gamma^\mu V_{IJ} d^J_L. \tag{4.255}$$

The electromagnetic current is vector-like:

$$J^\mu_{\rm em} = \sum_f Q_f \bar{f}\gamma^\mu f. \tag{4.256}$$

For one family,

$$J^\mu_{\rm em} = -\bar{e}\gamma^\mu e + \frac{2}{3}\bar{u}\gamma^\mu u - \frac{1}{3}\bar{d}\gamma^\mu d + \cdots, \tag{4.257}$$

where the dots indicate the remaining families.

The neutral weak current is chiral:

$$J^\mu_Z = \sum_f \bar{f}\gamma^\mu \left(g^f_L P_L + g^f_R P_R\right) f, \tag{4.258}$$

with

$$g^f_L = T^3_f - Q_f \sin^2\theta_W, \qquad g^f_R = -Q_f \sin^2\theta_W. \tag{4.259}$$

For one family this gives

$$\begin{aligned} J^\mu_Z = {} & \frac{1}{2}\bar{\nu}_L\gamma^\mu\nu_L + \left(-\frac{1}{2} + \sin^2\theta_W\right)\bar{e}_L\gamma^\mu e_L + \sin^2\theta_W \bar{e}_R\gamma^\mu e_R \\ & + \left(\frac{1}{2} - \frac{2}{3}\sin^2\theta_W\right)\bar{u}_L\gamma^\mu u_L - \frac{2}{3}\sin^2\theta_W \bar{u}_R\gamma^\mu u_R \\ & + \left(-\frac{1}{2} + \frac{1}{3}\sin^2\theta_W\right)\bar{d}_L\gamma^\mu d_L + \frac{1}{3}\sin^2\theta_W \bar{d}_R\gamma^\mu d_R. \end{aligned} \tag{4.260}$$

This expression is long, but its logic is simple: for each fermion, insert its T^3 and Q into Eq. (4.259).

One direct experimental consequence is seen in

$$e^+e^- \longrightarrow \gamma^*, Z^* \longrightarrow f\bar{f}. \tag{4.261}$$

The amplitude has a photon part and a Z-boson part:

$$\mathcal{M} = \mathcal{M}_\gamma + \mathcal{M}_Z. \tag{4.262}$$

The photon propagator is proportional to $1/s$, while the Z-exchange piece contains the resonance factor

$$\frac{1}{s - m_Z^2 + i m_Z \Gamma_Z}. \tag{4.263}$$

Near $s = m_Z^2$, the Z contribution dominates. Since $g^f_L \neq g^f_R$, the Z boson couples differently to left- and right-handed fermions. A useful measure of this difference is

$$A_f = \frac{\Gamma(Z \to f_L\bar{f}_R) - \Gamma(Z \to f_R\bar{f}_L)}{\Gamma(Z \to f_L\bar{f}_R) + \Gamma(Z \to f_R\bar{f}_L)} = \frac{(g^f_L)^2 - (g^f_R)^2}{(g^f_L)^2 + (g^f_R)^2}. \tag{4.264}$$

For a charged lepton,

$$g^\ell_L = -\frac{1}{2} + \sin^2\theta_W, \qquad g^\ell_R = \sin^2\theta_W. \tag{4.265}$$

Therefore

$$A_\ell = \frac{\left(-\frac{1}{2} + \sin^2\theta_W\right)^2 - \left(\sin^2\theta_W\right)^2}{\left(-\frac{1}{2} + \sin^2\theta_W\right)^2 + \left(\sin^2\theta_W\right)^2} \neq 0. \tag{4.266}$$

A nonzero asymmetry is a direct sign that the weak neutral current violates parity.

4.4 Masses from the Higgs field

Now substitute Eq. (4.247) into the Yukawa interactions. For charged leptons,

$$\begin{aligned}
-(Y_e)_{IJ}\bar{L}_L^I\Phi e_R^J + \text{h.c.} &= -(Y_e)_{IJ}\frac{v+h}{\sqrt{2}}\bar{e}_L^I e_R^J + \text{h.c.} \\
&= -(M_e)_{IJ}\bar{e}_L^I e_R^J - \frac{h}{v}(M_e)_{IJ}\bar{e}_L^I e_R^J + \text{h.c.},
\end{aligned} \tag{4.267}$$

where

$$M_e = \frac{v}{\sqrt{2}}Y_e. \tag{4.268}$$

Similarly,

$$M_d = \frac{v}{\sqrt{2}}Y_d, \qquad M_u = \frac{v}{\sqrt{2}}Y_u. \tag{4.269}$$

The Yukawa matrices are not generally diagonal. The physical masses are found by unitary rotations of the left- and right-handed fermion fields. For charged leptons,

$$U_{eL}^\dagger M_e U_{eR} = \text{diag}(m_e, m_\mu, m_\tau). \tag{4.270}$$

The quark mass matrices are diagonalized in the same way. The mismatch between the left-handed rotations for up-type and down-type quarks produces the CKM matrix in the charged current.

In the minimal Standard Model field content listed above, there is no right-handed neutrino. Hence there is no renormalizable Dirac Yukawa mass term for neutrinos, and the minimal theory predicts massless neutrinos. Neutrino oscillations show that this cannot be the full story.

A simple extension is to add right-handed neutrinos

$$\nu_R^I = (\nu_{eR}, \nu_{\mu R}, \nu_{\tau R}), \qquad \nu_R^I \sim (\mathbf{1}, \mathbf{1})_0\,. \tag{4.271}$$

They are singlets under all Standard Model gauge groups, so they are called sterile neutrinos. Once they are included, the Dirac neutrino Yukawa interaction is allowed:

$$\mathcal{L}_{\nu\text{Yuk}} = -(Y_\nu)_{IJ}\bar{L}_L^I\widetilde{\Phi}\nu_R^J + \text{h.c.} \tag{4.272}$$

Using explicit $SU(2)$ indices,

$$\bar{L}_L\widetilde{\Phi} = \epsilon_{ij}\Phi^{*i}\bar{L}_L^j. \tag{4.273}$$

After electroweak symmetry breaking,

$$\begin{aligned}
\mathcal{L}_{\nu\text{Yuk}} &= -(Y_\nu)_{IJ}\frac{v+h}{\sqrt{2}}\bar{\nu}_L^I\nu_R^J + \text{h.c.}, \\
M_\nu &= \frac{v}{\sqrt{2}}Y_\nu.
\end{aligned} \tag{4.274}$$

Because ν_R is a complete gauge singlet, a Majorana mass term is also allowed:

$$\mathcal{L}_{M_R} = -\frac{1}{2}(M_R)_{IJ}\overline{\nu_R^{cI}}\nu_R^J + \text{h.c.} \tag{4.275}$$

This is the starting point of the seesaw mechanism. The important lesson here is that neutrino masses require physics beyond the minimal Standard Model field content, even though they can be incorporated in a gauge-invariant way.

5. Quark Mixing and the CKM Matrix

The goal of this section is to explain why the Standard Model contains a mixing matrix in the quark charged current. The short version is simple: the weak interaction is written in a basis fixed by the gauge group, while particle masses are found by diagonalizing the Yukawa matrices. These two bases do not have to be the same. Their mismatch is the Cabibbo-Kobayashi-Maskawa matrix, usually called the CKM matrix.

5.1 Yukawa Couplings Before Electroweak Symmetry Breaking

For each generation $I = 1, 2, 3$, the Standard Model quarks are

$$Q_L^I = \begin{pmatrix} u_L^I \\ d_L^I \end{pmatrix}, \qquad u_R^I, \qquad d_R^I. \tag{4.276}$$

Here Q_L^I is an $SU(2)_L$ doublet, while u_R^I and d_R^I are $SU(2)_L$ singlets. The index I labels families. Before we diagonalize any mass matrix, it is better to call these fields *weak-interaction eigenstates*, not mass eigenstates.

The Higgs doublet is

$$\phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}, \qquad \widetilde{\phi} = \mathrm{i}\sigma^2 \phi^* = \begin{pmatrix} \phi^{0*} \\ -\phi^- \end{pmatrix}. \tag{4.277}$$

The object $\widetilde{\phi}$ is needed because the up-type quark Yukawa term must be made gauge invariant with the conjugate Higgs doublet. With generation indices shown explicitly, the quark Yukawa Lagrangian is

$$\mathcal{L}_Y^q = -\overline{Q}_L^I (Y_d)_{IJ} \phi \, d_R^J - \overline{Q}_L^I (Y_u)_{IJ} \widetilde{\phi} \, u_R^J + \text{h.c.} \tag{4.278}$$

The two matrices Y_u and Y_d are arbitrary complex 3×3 matrices. They are not assumed to be diagonal. Therefore the fields (u^I, d^I) appearing here are generally not the physical quarks (u, c, t, d, s, b).

For leptons, the analogous charged-lepton Yukawa term is

$$\mathcal{L}_Y^e = -\overline{L}_L^I (Y_e)_{IJ} \phi \, e_R^J + \text{h.c.} \tag{4.279}$$

where

$$L_L^I = \begin{pmatrix} \nu_L^I \\ e_L^I \end{pmatrix}. \tag{4.280}$$

The minimal Standard Model has no right-handed neutrino field and therefore no neutrino Yukawa mass term. If right-handed neutrinos ν_R^I are added, then one may also write

$$\mathcal{L}_Y^\nu = -\overline{L}_L^I (Y_\nu)_{IJ} \widetilde{\phi} \, \nu_R^J + \text{h.c.}, \tag{4.281}$$

but this is an extension of the minimal Standard Model. This distinction is important: quark mixing is already present in the minimal Standard Model, while neutrino mixing requires neutrino masses.

5.2 Mass Matrices After the Higgs Gets a Vacuum Expectation Value

In unitary gauge the Higgs field may be written as

$$\phi(x) = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}, \qquad \widetilde{\phi}(x) = \frac{1}{\sqrt{2}}\begin{pmatrix} v + h(x) \\ 0 \end{pmatrix}. \tag{4.282}$$

Substituting this into the Yukawa Lagrangian gives the quark mass terms

$$\mathcal{L}^q_{\text{mass}} = -\overline{d}^I_L (M_d)_{IJ} d^J_R - \overline{u}^I_L (M_u)_{IJ} u^J_R + \text{h.c.}, \tag{4.283}$$
$$M_d = \frac{v}{\sqrt{2}} Y_d, \qquad M_u = \frac{v}{\sqrt{2}} Y_u. \tag{4.284}$$

The matrices M_u and M_d are complex 3×3 matrices. A generic complex matrix is not diagonal and not Hermitian. Therefore we do not diagonalize it by a single unitary transformation. Instead, we use a bi-unitary transformation, also called singular value decomposition.

For any complex matrix M, there exist unitary matrices V_L and V_R such that

$$V_L^\dagger M V_R = D, \tag{4.285}$$

where D is diagonal with non-negative real entries. In the quark sector we choose

$$V_{uL}^\dagger M_u V_{uR} = D_u = \text{diag}(m_u, m_c, m_t), \tag{4.286}$$
$$V_{dL}^\dagger M_d V_{dR} = D_d = \text{diag}(m_d, m_s, m_b). \tag{4.287}$$

The positive numbers on the diagonal are the physical quark masses.

The corresponding change of basis is

$$u^I_L = (V_{uL})_{Ii} u^i_L, \qquad u^I_R = (V_{uR})_{Ii} u^i_R, \tag{4.288}$$
$$d^I_L = (V_{dL})_{Ij} d^j_L, \qquad d^I_R = (V_{dR})_{Ij} d^j_R. \tag{4.289}$$

Uppercase indices I, J label the original weak basis, while lowercase indices i, j label the mass basis. To see explicitly why this works, look at the up-quark mass term:

$$-\overline{u}^I_L (M_u)_{IJ} u^J_R = -\overline{u}^i_L (V_{uL}^\dagger)_{iI} (M_u)_{IJ} (V_{uR})_{Jj} u^j_R \tag{4.290}$$
$$= -\overline{u}^i_L (D_u)_{ij} u^j_R \tag{4.291}$$
$$= -m_u \overline{u}_L u_R - m_c \overline{c}_L c_R - m_t \overline{t}_L t_R. \tag{4.292}$$

Adding the Hermitian conjugate gives the usual Dirac mass terms:

$$-m_i \overline{q}_{Li} q_{Ri} - m_i \overline{q}_{Ri} q_{Li} = -m_i \overline{q}_i q_i. \tag{4.293}$$

5.3 Why Most Gauge Interactions Do Not Notice the Basis Change

The kinetic terms are invariant under these constant unitary rotations. For example,

$$\mathrm{i}\overline{u}^I_L \gamma^\mu \partial_\mu u^I_L = \mathrm{i}\overline{u}^i_L (V_{uL}^\dagger)_{iI} \gamma^\mu \partial_\mu (V_{uL})_{Ij} u^j_L \tag{4.294}$$
$$= \mathrm{i}\overline{u}^i_L (V_{uL}^\dagger V_{uL})_{ij} \gamma^\mu \partial_\mu u^j_L \tag{4.295}$$
$$= \mathrm{i}\overline{u}^i_L \gamma^\mu \partial_\mu u^i_L. \tag{4.296}$$

The derivative does not act on V_{uL}, because this is a constant matrix in generation space, not a spacetime-dependent gauge transformation.

Neutral gauge currents are also unchanged. A typical neutral up-quark current becomes

$$\overline{u}^I_L \gamma^\mu u^I_L = \overline{u}^i_L (V^\dagger_{uL} V_{uL})_{ij} \gamma^\mu u^j_L \tag{4.297}$$
$$= \overline{u}^i_L \gamma^\mu u^i_L. \tag{4.298}$$

The same argument applies to the photon, gluon, and Z-boson neutral currents. This is why tree-level flavor-changing neutral currents are absent in the Standard Model. The important exception is the charged weak current.

5.4 The Charged Current and the CKM Matrix

In the weak basis, the charged-current interaction is diagonal in family space:

$$\mathcal{L}_{\text{cc}} = -\frac{g}{\sqrt{2}} \overline{u}^I_L \gamma^\mu d^I_L W^+_\mu - \frac{g}{\sqrt{2}} \overline{d}^I_L \gamma^\mu u^I_L W^-_\mu. \tag{4.299}$$

Now substitute the mass-basis rotations:

$$\overline{u}^I_L \gamma^\mu d^I_L = \overline{u}^i_L (V^\dagger_{uL})_{iI} \gamma^\mu (V_{dL})_{Ij} d^j_L \tag{4.300}$$
$$= \overline{u}^i_L \gamma^\mu (V^\dagger_{uL} V_{dL})_{ij} d^j_L. \tag{4.301}$$

This motivates the definition

$$V_{\text{CKM}} \equiv V^\dagger_{uL} V_{dL}. \tag{4.302}$$

The charged-current Lagrangian in the mass basis is therefore

$$\mathcal{L}_{\text{cc}} = -\frac{g}{\sqrt{2}} \overline{u}^i_L \gamma^\mu (V_{\text{CKM}})_{ij} d^j_L W^+_\mu + \text{h.c.} \tag{4.303}$$

$$= -\frac{g}{\sqrt{2}} \begin{pmatrix} \overline{u}_L & \overline{c}_L & \overline{t}_L \end{pmatrix} \gamma^\mu V_{\text{CKM}} \begin{pmatrix} d_L \\ s_L \\ b_L \end{pmatrix} W^+_\mu + \text{h.c.} \tag{4.304}$$

Written out as a matrix,

$$V_{\text{CKM}} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix}. \tag{4.305}$$

Each entry tells us the amplitude for an up-type quark to couple to a down-type quark through a W boson. For example, V_{us} controls the strength of $u \leftrightarrow s$ charged-current transitions.

Because V_{uL} and V_{dL} are unitary, the CKM matrix is unitary:

$$V_{\text{CKM}} V^\dagger_{\text{CKM}} = V^\dagger_{uL} V_{dL} V^\dagger_{dL} V_{uL} = \mathbb{1}, \tag{4.306}$$
$$V^\dagger_{\text{CKM}} V_{\text{CKM}} = V^\dagger_{dL} V_{uL} V^\dagger_{uL} V_{dL} = \mathbb{1}. \tag{4.307}$$

5.5 Discrete Symmetries and Why the Weak Current Is Special

The electromagnetic and strong interactions are invariant under C, P, and T, and hence also under CPT. The weak charged current is different because it only contains left-handed fermion fields:

$$J^{+\mu}_W = \overline{u}^i_L \gamma^\mu (V_{\text{CKM}})_{ij} d^j_L. \tag{4.308}$$

Parity exchanges left-handed and right-handed chiral fields:

$$P\psi_L(t,\mathbf{x})P^{-1} = \eta_P\gamma^0\psi_R(t,-\mathbf{x}), \tag{4.309}$$

where η_P is an irrelevant phase. Therefore a current made only from left-handed fields cannot be invariant under parity by itself. This is the origin of maximal parity violation in the charged weak interaction.

Charge conjugation changes particles into antiparticles. With a charge conjugation matrix C satisfying

$$C^{-1}\gamma^\mu C = -(\gamma^\mu)^T, \tag{4.310}$$

one defines

$$\psi^C = C\overline{\psi}^T. \tag{4.311}$$

The charged weak current is not invariant under C alone either. The combined transformation CP is more subtle: a left-handed particle is mapped into the corresponding right-handed antiparticle, which can still participate in weak interactions. Thus the chiral structure alone does not force CP violation.

The CKM matrix is where CP violation enters. Under CP, the charged-current coupling is complex conjugated:

$$V_{\text{CKM}} \longrightarrow V_{\text{CKM}}^*. \tag{4.312}$$

If all phases in V_{CKM} can be removed by redefining quark fields, then CP is conserved. If one complex phase remains, then CP is violated.

5.6 Counting the Physical Parameters in the CKM Matrix

A general 3×3 unitary matrix has nine real parameters. It is useful to think of them as three rotation angles and six phases:

$$9 = 3 \text{ angles} + 6 \text{ phases}. \tag{4.313}$$

However, not all six phases are physical. The six quark mass eigenstate fields can be rephased:

$$u_i \longrightarrow \mathrm{e}^{\mathrm{i}\alpha_i}u_i, \qquad\qquad d_j \longrightarrow \mathrm{e}^{\mathrm{i}\beta_j}d_j. \tag{4.314}$$

Under this change,

$$(V_{\text{CKM}})_{ij} \longrightarrow \mathrm{e}^{-\mathrm{i}\alpha_i}(V_{\text{CKM}})_{ij}\mathrm{e}^{\mathrm{i}\beta_j}. \tag{4.315}$$

There are six phases α_i, β_j, but one common phase changes nothing:

$$\alpha_1=\alpha_2=\alpha_3=\beta_1=\beta_2=\beta_3 \quad\Longrightarrow\quad V_{\text{CKM}} \text{ unchanged}. \tag{4.316}$$

Therefore only five phases can be removed. The physical CKM matrix has

$$3 \text{ angles} + (6-5) \text{ phase} = 3 \text{ angles} + 1 \text{ CP-violating phase}. \tag{4.317}$$

For n generations, the general result is

$$N_{\text{angles}} = \frac{n(n-1)}{2}, \tag{4.318}$$

$$N_{\text{phases}} = \frac{(n-1)(n-2)}{2}. \tag{4.319}$$

For $n=2$, there is one mixing angle and no physical complex phase. This is why two generations are not enough for CKM-type CP violation. With three generations, one irreducible complex phase remains.

5.7 Useful Parameterizations

The standard parameterization writes the CKM matrix in terms of three angles $\theta_{12}, \theta_{23}, \theta_{13}$ and one phase δ. Using $s_{ij} = \sin\theta_{ij}$ and $c_{ij} = \cos\theta_{ij}$,

$$V_{\text{CKM}} = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}\mathrm{e}^{-\mathrm{i}\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}\mathrm{e}^{\mathrm{i}\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}\mathrm{e}^{\mathrm{i}\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}\mathrm{e}^{\mathrm{i}\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}\mathrm{e}^{\mathrm{i}\delta} & c_{23}c_{13} \end{pmatrix}. \tag{4.320}$$

This form makes the single physical phase explicit. If $\delta = 0$ or $\delta = \pi$, then the matrix can be chosen real and there is no CKM-type CP violation.

Phenomenologically, the CKM matrix is close to the identity matrix. The Wolfenstein parameterization makes this hierarchy transparent:

$$V_{\text{CKM}} = \begin{pmatrix} 1 - \frac{\lambda^2}{2} & \lambda & A\lambda^3(\rho - \mathrm{i}\eta) \\ -\lambda & 1 - \frac{\lambda^2}{2} & A\lambda^2 \\ A\lambda^3(1 - \rho - \mathrm{i}\eta) & -A\lambda^2 & 1 \end{pmatrix} + O(\lambda^4), \qquad \lambda \simeq 0.22. \tag{4.321}$$

Here λ measures the Cabibbo mixing between the first two generations, A controls the size of mixing with the third generation, and η is responsible for CP violation.

5.8 Unitarity Triangles and the Jarlskog Invariant

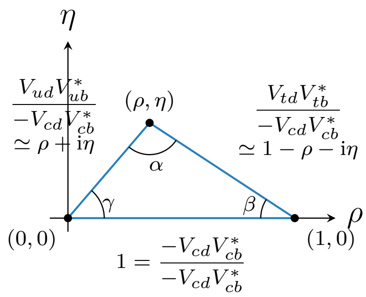
The unitarity of V_{CKM} gives orthogonality relations among different rows and columns. One important relation is

$$V_{ud}V_{ub}^* + V_{cd}V_{cb}^* + V_{td}V_{tb}^* = 0. \tag{4.322}$$

This is the sum of three complex numbers. Since the sum is zero, the three numbers form a triangle in the complex plane. Dividing by $-V_{cd}V_{cb}^*$ gives the conventional normalized triangle:

$$\frac{V_{ud}V_{ub}^*}{-V_{cd}V_{cb}^*} + \frac{V_{td}V_{tb}^*}{-V_{cd}V_{cb}^*} = 1. \tag{4.323}$$

At leading order in the Wolfenstein expansion, the apex is approximately (ρ, η):



The angles α, β, γ are measured in many flavor-physics processes. The fact that the triangle has nonzero area is equivalent to the presence of CP violation.

A basis-independent measure of CKM CP violation is the Jarlskog invariant:

$$J = \operatorname{Im}\left(V_{ij}V_{kl}V_{il}^*V_{kj}^*\right), \qquad i \neq k, \quad j \neq l. \tag{4.324}$$

For a unitary 3×3 CKM matrix, every nonzero choice gives the same magnitude of J. In the Wolfenstein expansion,

$$J = A^2\lambda^6\eta + O(\lambda^8). \tag{4.325}$$

Thus $J = 0$ if the physical phase vanishes, if one of the mixing angles vanishes, or if any two quarks with the same charge are exactly degenerate in mass. A compact invariant way to state the last point is

$$\operatorname{Im}\det\left[M_u M_u^\dagger, M_d M_d^\dagger\right] \propto J \prod_{i<j}(m_{u_i}^2 - m_{u_j}^2) \prod_{k<l}(m_{d_k}^2 - m_{d_l}^2). \tag{4.326}$$

This formula explains why CP violation needs both a complex CKM phase and a genuinely three-generation mass spectrum.

5.9 Parameter Count in the Flavor Sector

If neutrino masses are ignored, the usual Standard Model parameter count can be summarized as

$$\text{gauge couplings}: 3, \tag{4.327}$$
$$\text{Higgs potential parameters}: 2 \quad (v, \lambda), \tag{4.328}$$
$$\text{quark masses}: 6, \tag{4.329}$$
$$\text{charged-lepton masses}: 3, \tag{4.330}$$
$$\text{CKM parameters}: 4. \tag{4.331}$$

This gives

$$3 + 2 + 6 + 3 + 4 = 18 \tag{4.332}$$

parameters, if the QCD theta angle is not included in the count.

If neutrinos are massive, the lepton sector contains additional physical parameters. For three light Majorana neutrinos one commonly adds three neutrino masses, three PMNS mixing angles, one Dirac phase, and two Majorana phases. If one counts only oscillation physics, the two Majorana phases do not appear, and the extra oscillation-relevant parameters are

$$3 \text{ neutrino masses} + 4 \text{ PMNS mixing parameters} = 7. \tag{4.333}$$

The moral is that flavor physics is mostly the study of how these mass matrices are diagonalized and how the mismatch between different diagonalizations appears in weak interactions.